

Proposition 5.10. *Suppose that $(U_*, \lambda_*) \in V(r, n_1) \times \cdots \times V(r, n_s) \times \text{OB}(r, n_{s+1}) \times \cdots \times \text{OB}(r, n_k) \times \mathbb{R}_*^r$. Then (U_*, λ_*) is a critical point of g if and only if it is a KKT point of (32).*

The following fact is also straightforward to verify.

Lemma 5.11 (Local Diffeomorphism). *[49] The set $W_r := V(r, n_1) \times \cdots \times V(r, n_s) \times \text{OB}(r, n_{s+1}) \times \cdots \times \text{OB}(r, n_k) \times \mathbb{R}_*^r$ is a smooth manifold and is locally diffeomorphic to the manifold $\psi(W_r)$, where ψ is the map defined in (23).*

Theorem 5.12 (Generic Linear Convergence Rate). *Let $s \geq 1$ and let $\{U_{[p]}\}$ be a sequence generated by the iAPD-ALS algorithm for a generic tensor $\mathcal{A} \in \mathbb{R}^{n_1} \otimes \cdots \otimes \mathbb{R}^{n_k}$. If $s = 1$, assume in addition that*

$$\sum_{i'=2}^k n_{i'} \geq n_1 + k - 1, \quad \sum_{i'=2}^k n_{i'} \geq n_{i''} + k, \quad 2 \leq i'' \leq k.$$

The sequence $\{U_{[p]}\}$ converges R -linearly to a KKT point of (32).

Proof. It suffices to verify conditions a)–d) in Theorem 5.5. It follows from Lemma 5.8 that there is some $t \leq r$ such that $(U_{[p]}, \lambda_{[p]}) \in V(t, n_1) \times \cdots \times V(t, n_s) \times B(t, n_{s+1}) \times \cdots \times B(t, n_k) \times \mathbb{R}_*^t$ for sufficiently large p . According to Lemma 5.11, W_t is locally diffeomorphic to $\psi(W_t)$. Hence a) is satisfied. Moreover, we already see that b) holds in the proof of Proposition 4.4. By Proposition 5.6, the sequence $\{(U_{[p]}, \lambda_{[p]})\}$ globally converges to a KKT point (U_*, λ_*) of (32) and this verifies c). Lastly, Proposition 5.7 and Lemma 5.9 ensure that d) is satisfied. $\square$

Note that by Corollary 4.5, the requirement for (25) and (26) can be removed if $s = 1, k \geq 4$ and $n_1 = \cdots = n_k$. Hence we obtain the following.

Theorem 5.13 (Square Tensors). *Assume that $k \geq 4$ and $n_1 = \cdots = n_k \geq 2$. For a generic $\mathcal{A}$, the sequence $\{U_{[p]}\}$ generated by Algorithm 1 converges R -linearly to a KKT point of (32).*

We remark that the sublinear convergence rate can be established for the general case [49], while in Theorem 5.12 linear convergence rate is given for a generic case. This can be interpreted via the fact that the local branches of the solution mapping for the best low rank partially orthogonal tensor approximation problem have local Lipschitz continuity whenever the nondegeneracy of the converged KKT point holds [12]. However, the nondegeneracy only holds generically. Equivalently, for general polynomial systems, only local Hölderian error bounds can hold [34]. The explicit examples given in [16, 17] present a witness for this phenomenon when $r = 1$.

6. CONCLUSIONS

In this paper, we study the non-linear least squares (NLS) problem associated with a smooth variety M and a polynomial map $\psi : M \rightarrow \mathbb{R}^n$. This model encompasses a wide range of practical problems, including low rank (structured) matrix approximation problems and low rank (structured) tensor approximation problems. Under mild assumptions, we prove that critical points of the NLS problem are all mapped into the smooth locus of $\psi(M)$. One significant implication of this result is that, when designing and analyzing algorithms for the NLS problem, misbehaved points are guaranteed to be avoided. As a consequence, we obtain a linear convergence rate of an iterative sequence generated by algorithms for the NLS problem. In particular, we apply our general framework to establish the generic linear convergence rate of the iAPD-ALS algorithm for the low rank partially orthogonal tensor approximation problem, without any further assumption.

Given that numerous problems can be recast as the NLS problem for different choices of M and ψ , our framework is potentially applicable to the convergence analysis of numerical algorithms for them. It is possible to establish an unconditional convergence result analogous to Theorem 5.13. We also want to mention that our framework heavily relies on the geometry of M and ψ . Hence, the framework is applicable if the geometry of (M, ψ) is well-understood. Of particular interest is the case where $\psi(M)$ is a variety of structured matrices or tensors, since such varieties are already extensively studied in the context of computational algebraic geometry [13, 31, 35].

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